ISI MSQE PEA 2024 PYQ Practice Booklet

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Question 1

If $f : \mathbb{R} \to \mathbb{R}$ is a differentiable function at $a \in \mathbb{R}$ such that $f'(a) = af(a)$, then what is

$$\lim_{x \to a} \frac{xf(a) - af(x)}{x - a}\,?$$

(A) $af(a)$

(B) $f(a)$

(C) $(1 - a^2)f(a)$

(D) None of the previous options

Question 2

Suppose S_n is defined as follows for every positive integer $n \geq 2$:

$$S_n = \left(1 - \tfrac{1}{2^2}\right)\left(1 - \tfrac{1}{3^2}\right)\cdots\left(1 - \tfrac{1}{n^2}\right).$$

The value of $\lim\limits_{n \to \infty} S_n$ is

(A) 0

(B) $\frac{1}{2}$

(C) 1

(D) ∞

Question 3

Suppose

$$\lim_{x \to 0} \frac{e^{a_1 x} - 1}{a_2 x^2 + a_3 x} = 1,$$

where a_1, a_2, a_3 are given real numbers. Then it is necessarily true that

(A) $a_1 = a_2 = a_3 = 1$

(B) $a_1 = a_3 \neq 0$

(C) $a_2 = 0$

(D) $a_2 + a_3 \neq 0$

Question 4

Let $f : \mathbb{R} \to \mathbb{R}$ be defined as

$$f(x) = \begin{cases} cx^2 + ax + b, & x < 0, \\ bx^2 + cx + a, & 0 \leq x < 2, \\ ax^2 + bx + c, & x \geq 2, \end{cases}$$

where a, b, c are positive real numbers. Which of the following statements is correct, under the assumption that f is continuous?

(A) f is continuous for all values of a, b, c

(B) f is continuous iff $a - b = b - c$

(C) f is continuous iff $a = b$ and $c = 2a$

(D) f is continuous iff $a = b = c$

Question 5

Consider $f : [0, 1] \to [0, 1]$ such that $f(x) = \dfrac{x}{2 - x}$. Which of the following statements is *incorrect*?

(A) $f(0) = 0$ and $f(1) = 1$

(B) $f(1 - f(x)) = 1 - x$

(C) f is strictly concave in the interval $(0, 1)$

(D) f is strictly increasing in the interval $(0, 1)$

Question 6

Let $f : \mathbb{R} \to \mathbb{R}$ be the function $f(x) = |x| + x^2 \ \forall x \in \mathbb{R}$. Which of the following statements about f is correct?

(A) f is differentiable

(B) f is concave but not differentiable

(C) f is convex but not differentiable

(D) f is discontinuous

Question 7

$\int x^3 e^{x^2}\, dx$ equals

(A) $\dfrac{x(x-1)}{2}\, e^{x^2}$

(B) $\dfrac{x^2-1}{2}\, e^{x^2}$

(C) $\dfrac{x(x+1)}{2}\, e^{x^2}$

(D) $\dfrac{x^2+1}{2}\, e^{x^2}$

Question 8

Let A be a 3×3 matrix having eigenvalues $2, 7, 5$. What is the determinant of $A + 2I$?

(A) 252

(B) 70

(C) 420

(D) 84

Question 9

Let x and y be two column vectors of length 3 such that $\sum_i x_i y_i = 1$. What is the rank of xy^T?

(A) 0

(B) 1

(C) 2

(D) 3

Question 10

Let A be a 3×3 matrix such that $Ax = x$ for all column vectors x of length 3. Which of the following statements is correct?

(A) No such A exists

(B) $A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$

(C) $A = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}$

(D) A exists but is different from the matrices given in parts (B) and (C)

Question 11

Let $S = \{(x_1, 0) : x_1 \in \mathbb{R}\}$ and $T = \{(0, x_2) : x_2 \in \mathbb{R}\}$. Which of the following is correct?

(A) S, T and $S \cap T$ are vector spaces

(B) S, T and $S \cup T$ are vector spaces

(C) $S \cup T$ and $S \cap T$ are vector spaces

(D) Neither $S \cup T$ nor $S \cap T$ is a vector space

Question 12

In a chess tournament, there are both boys and girls. Each player plays with another player exactly once. If there are 45 games in total and exactly 15 of them feature only boys, how many games feature a boy and a girl?

(A) 6

(B) 15

(C) 20

(D) 24

Question 13

What is the number of possible arrangements of the letters of the word 'madam' such that the two 'a's never appear in consecutive positions?

(A) 30

(B) 24

(C) 18

(D) 12

Question 14

A monkey starts at $(0,0)$ on the xy-plane in period 1. From any position (x,y) in a period, the monkey can only jump to (a,b) in the next period, where $a \in \{x+1, x, x-1\}$ and $b \in \{y+1, y, y-1\}$. How many possible positions can the monkey be in period 2?

(A) 9

(B) 6

(C) 4

(D) 3

Question 15

In the previous question, suppose in every period the monkey can go to any of the possible positions in the next period with equal probability. Then what is the probability that the monkey is at a distance of more than 1 from $(0,0)$ in period 2?

(A) 0

(B) $\frac{16}{81}$

(C) $\frac{1}{3}$

(D) $\frac{4}{9}$

Question 16

For a given data set, let the least squares regression line be $y = 10 + 2x$. It is given that variance of x is 9 and variance of y is 81. What is the correlation coefficient between x and y?

(A) $\frac{1}{3}$

(B) $\frac{1}{2}$

(C) $\frac{2}{3}$

(D) $\frac{3}{4}$

Question 17

An urn contains 4 white, 6 red, and 5 black balls. 5 balls are randomly selected from the urn. Let X and Y denote respectively the number of white and black balls selected. Suppose $Y = 2$, i.e., 2 of the 5 selected balls are black. What is the probability that X takes the value 2?

(A) $\frac{3}{10}$

(B) $\frac{4}{10}$

(C) $\frac{3}{15}$

(D) $\frac{4}{15}$

Question 18

In answering an MCQ with 4 choices, a student either knows the answer (with probability $\frac{1}{4}$) or guesses (with probability $\frac{3}{4}$). A guess is correct with probability $\frac{1}{4}$. What is the probability that the student knew the answer, given that he answered correctly?

(A) $\frac{3}{7}$

(B) $\frac{4}{7}$

(C) $\frac{3}{4}$

(D) 1

Question 19

Let $X_1, X_2, \ldots, X_5$ be i.i.d. random variables with $E(X_i) = 0$ and $V(X_i) = 1$. What is the value of

$$E\left[\frac{(\sum_{i=1}^{5} X_i)^2}{5}\right]?$$

(A) 0

(B) 1

(C) -1

(D) None of the previous options

Question 20

Consider $X \sim \text{Bin}(n, p)$. Which of the following is *incorrect*?

(A) If np is an integer, then the mean and mode of this distribution coincide

(B) If n is even and $p = \frac{1}{2}$, then the median of this distribution is $n/2$

(C) If $P(X = k) = P(X = n - k)$ for every $k \in \{0, 1, \ldots, n\}$, then $p = \frac{1}{2}$

(D) If $(n + 1)p - 1$ is an integer, then this distribution is unimodal

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Question 21

Consider an economy with $y = Ak$ (per-capita form), saving rate $s \in (0,1)$, population growth rate $n > 0$ and depreciation rate $\delta \in (0,1)$. Assume parameters ensure positive long-run growth of y. Which of the following is *incorrect*?

(A) The growth rate of capital–labour ratio is $sA - (n + \delta)$

(B) If investment $I > \delta K$, then output grows at a positive rate

(C) The economy is always on the steady state growth path

(D) Increasing the savings rate s will not have any effect on the long-run growth rate of output per worker

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Question 22

Solow economy with $Y = K^{1/3}L^{2/3}$, no population growth or technological change, $\delta = 0.05$, $s = 0.2$. What is the steady-state capital–labour ratio?

(A) 6

(B) 8

(C) 2

(D) 4

Question 23

With the same Solow economy (no population growth, $\delta = 0.05$, $Y = K^{1/3}L^{2/3}$), a social planner wishes to maximize steady-state per-capita consumption. What savings rate s achieves this?

(A) $\frac{2}{3}$

(B) $\frac{1}{2}$

(C) $\frac{1}{3}$

(D) $\frac{1}{4}$

Question 24

A Solow economy has $y = k^{1/2}$, $\delta = 0$, labour grows at $n > 0$. If the steady-state value of k is 50 and the current value of k is 2, what is the current growth rate of per-capita output?

(A) $24n$

(B) $25n$

(C) 0.3

(D) None of the previous options

Question 25

Two countries A and B produce wheat and television respectively, with $Y_A = 200$ and $Y_B = 250$. The TV price is normalised to 1, the relative price of wheat is p. Let E_A (in wheat units) and E_B (in TV units) denote total expenditures. The income identity is $pY_A + Y_B = pE_A + E_B$. Initially $E_A = 100$; A reduces E_A to 70. Suppose 1/3 of each country's expenditure is on wheat. What happens to the equilibrium p?

(A) p stays the same

(B) p increases

(C) p decreases

(D) The effect is ambiguous

Question 26

Same setup as Q25, except now *each* country spends 2/3 of its expenditure on its *own* good and 1/3 on the foreign good. What is the impact on p when E_A falls from 100 to 70?

(A) p stays the same

(B) p increases

(C) p decreases

(D) The effect is ambiguous

Question 27

A consumer consumes two goods X and Y. It is observed that her consumption of X always falls when the price of X falls, *ceteris paribus*. Suppose now income rises, with prices of X and Y held constant. What happens to consumption of X?

(A) Consumption of X falls

(B) Consumption of X rises

(C) Consumption of X remains unchanged

(D) Indeterminate: consumption of X could rise, fall, or remain unchanged

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Question 28

A has a pond which yields f fish at $t = 1$ and nothing at $t = 2$. He consumes only fish. Storage is costless and undepreciated; no discounting. Utility in any period t is $u(c_t) = c_t^n$ with $0 < n < 1$. Optimal consumption?

(A) Any division is optimal

(B) $c_1 = c_2 = f/2$

(C) $c_1 = f,\ c_2 = 0$

(D) $c_1 = 0,\ c_2 = f$

Question 29

Country C has 10,000 identical farmers with cost $C(q_i) = 0.5q_i^2 + 4q_i + 100$, aggregate demand $Q = -10{,}000p + 400{,}000$. Current price support $p = 30$; government buys all unsold output at 30. The government proposes removing the support and giving each farmer an equal share of the saved expenditure as a lump-sum transfer. Each farmer must, however, pay 300 to open a bank account to receive the transfer. By how much does a farmer gain (or lose)?

(A) Rise by 8

(B) Rise by 4

(C) Fall by 4

(D) Fall by 8

Question 30

A consumer consumes three goods X, Y, Z over three periods. Prices and bundles:

$$\begin{aligned} &\text{Period 1: } p^1 = (2,3,3),\ x^1 = (3,1,7), \\ &\text{Period 2: } p^2 = (3,2,3),\ x^2 = (7,3,1), \\ &\text{Period 3: } p^3 = (3,3,2),\ x^3 = (1,7,3). \end{aligned}$$

Which of the following statements about her preferences is correct?

(A) Preferences are not complete

(B) Preferences are transitive though not necessarily complete

(C) Preferences are incomplete and intransitive

(D) Preferences are complete and transitive
